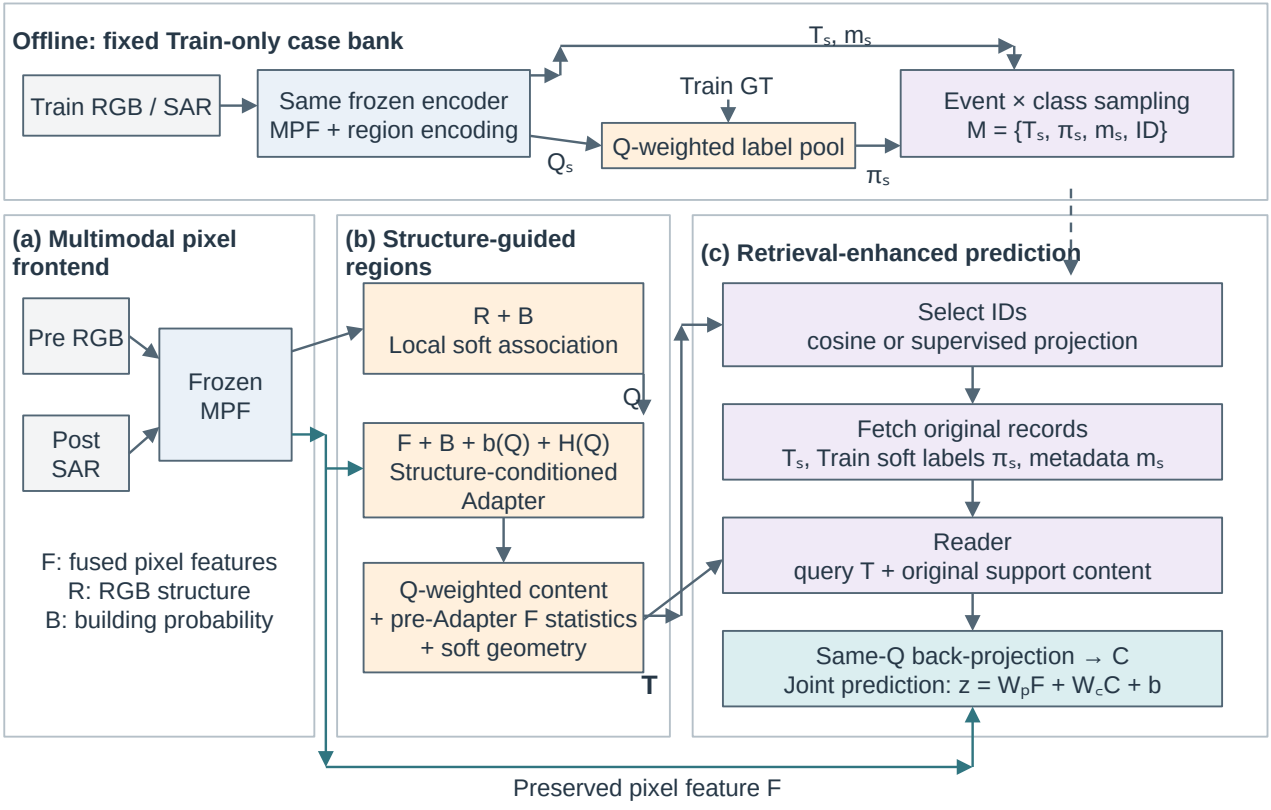
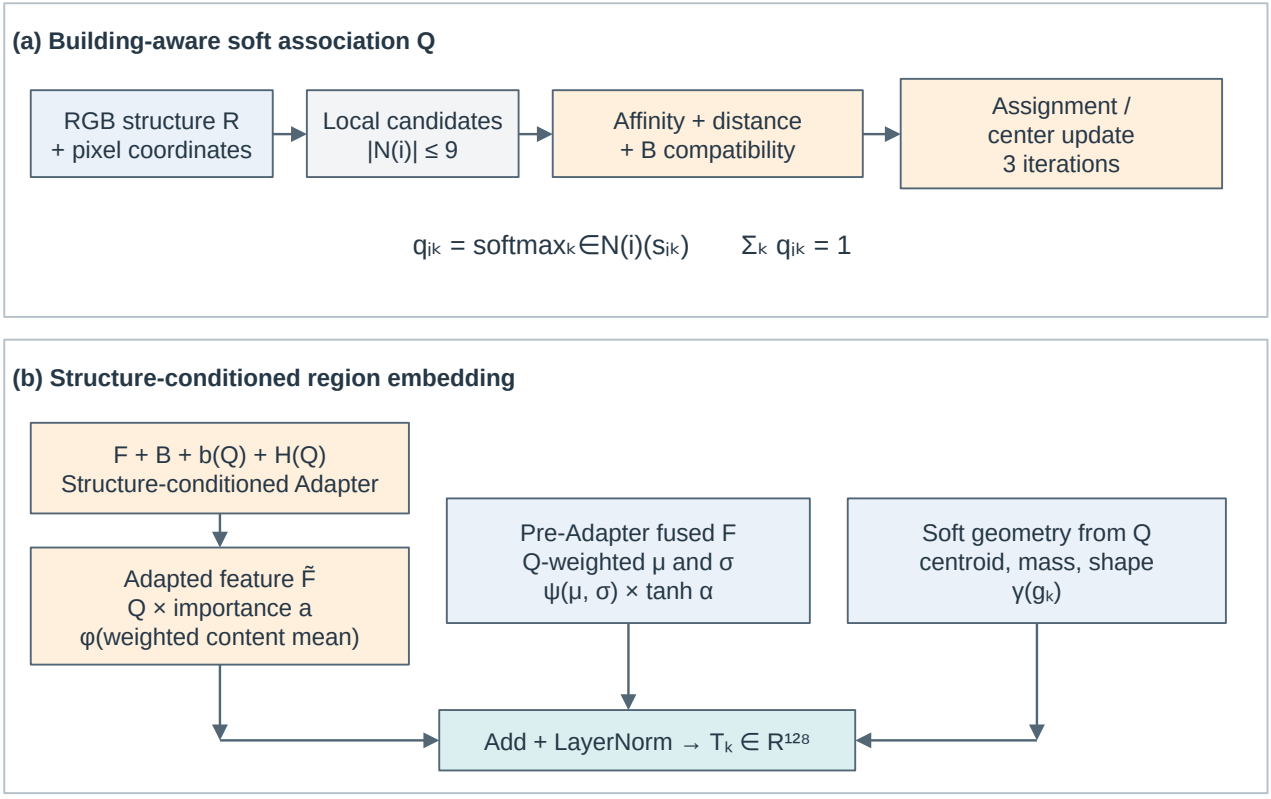
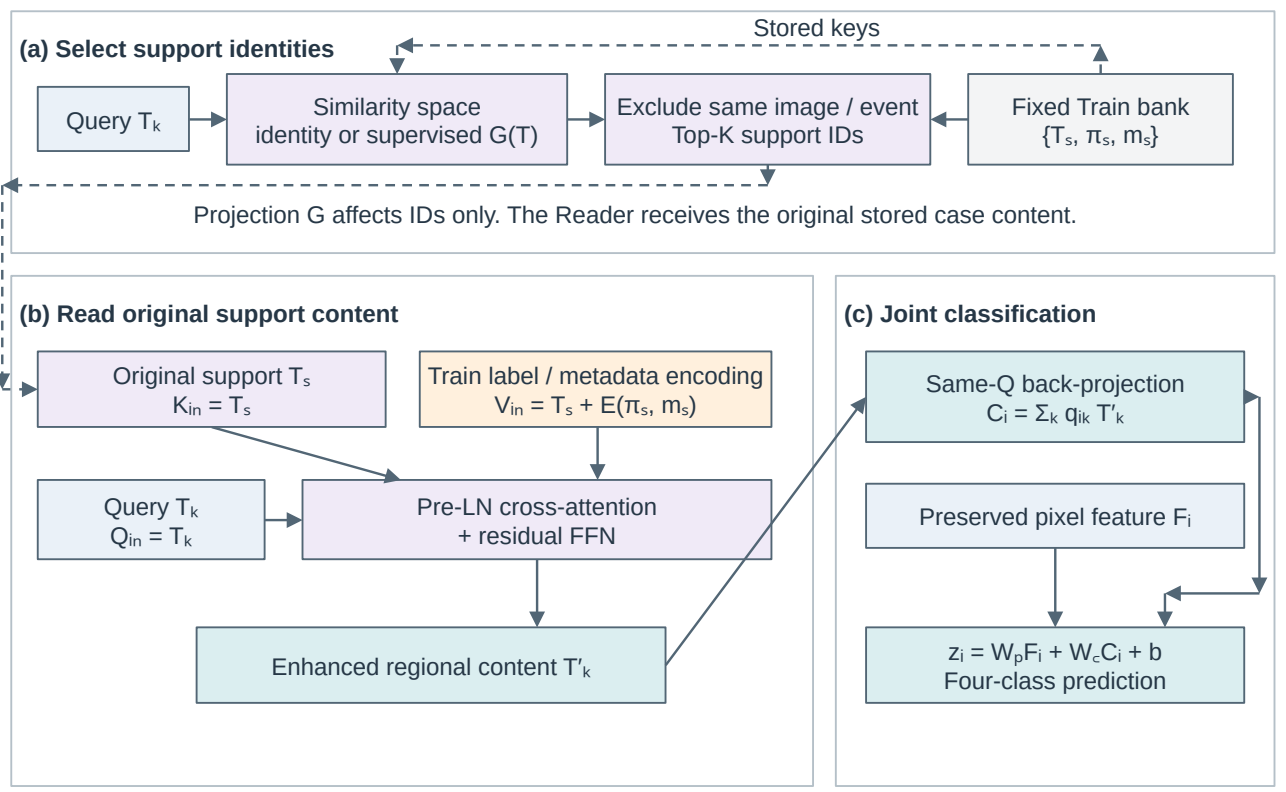


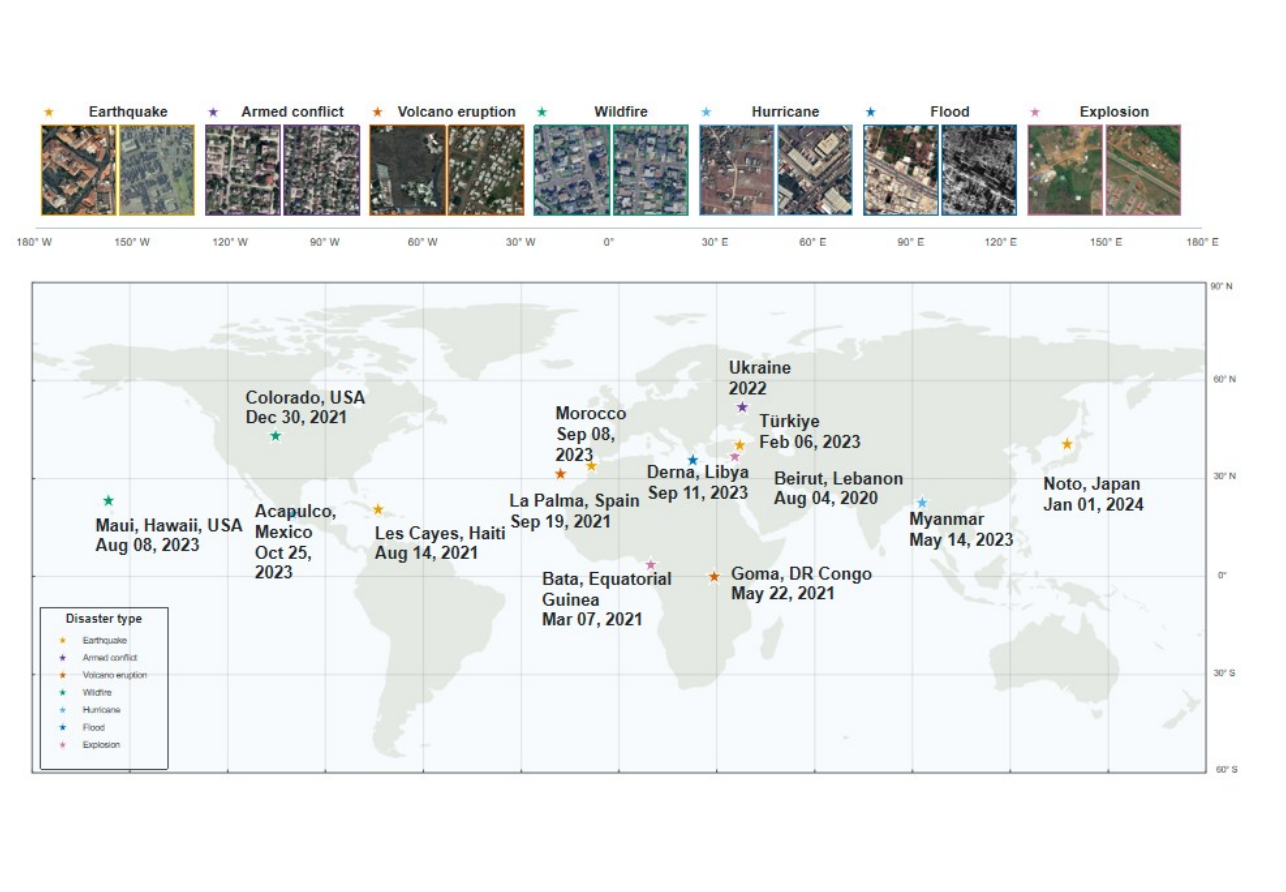
Solid arrows: feature flow. Dashed arrow: case-bank lookup. Val / Test GT is never an input.

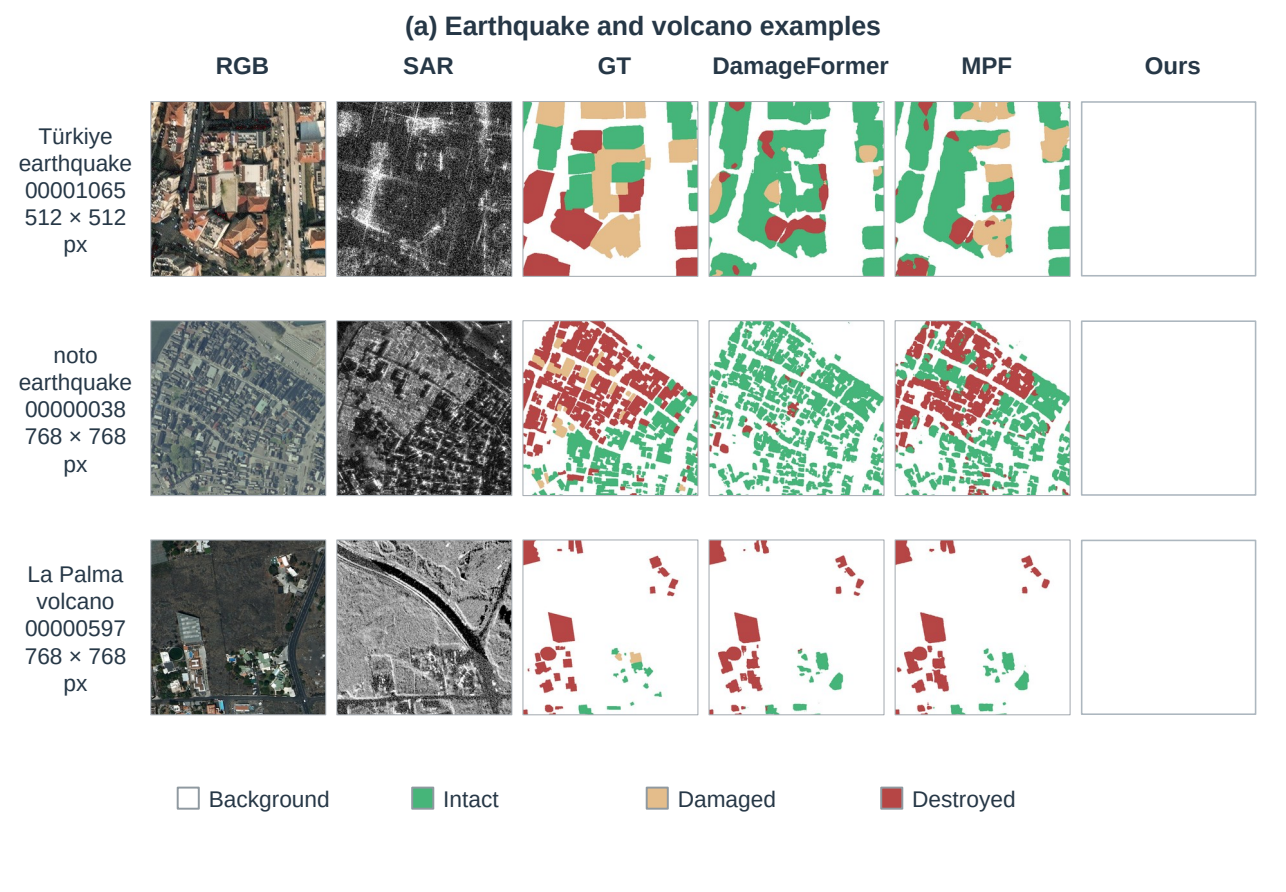


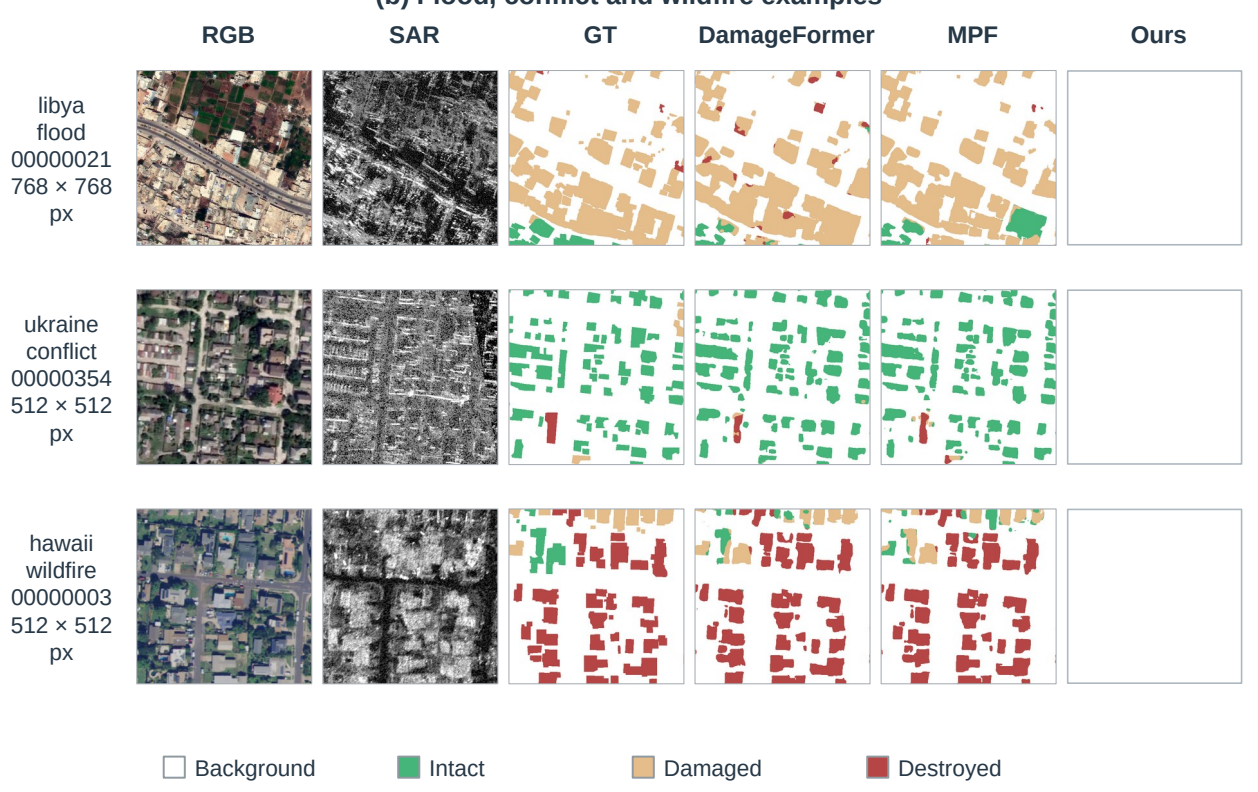
B is predicted building probability; b(Q) is assignment discrepancy; H(Q) is association entropy. Neither is a GT map.



Parametric-support control uses learned reference vectors instead of Train cases. It is an ablation, not a claim of memory gain.


Earthquake
Armed conflict
Volcano eruption
Wildfire
Hurricane
Flood
Explosion
180° W
150° W
120° W
90° W
60° W
30° W
0°
30° E
60° E
90° E
120° E
150° E
180° E
90° N
60° N
30° N
0°
30° S
60° S
Colorado, USA
Dec 30, 2021
Morocco
Sep 08,
2023
Ukraine
2022
Türkiye
Feb 06, 2023
Derna, Libya
Sep 11, 2023
Beirut, Lebanon
Aug 04, 2020
Noto, Japan
Jan 01, 2024
Maui, Hawaii, USA
Aug 08, 2023
Acapulco,
Mexico
Oct 25,
2023
Les Cayes, Haiti
Aug 14, 2021
La Palma, Spain
Sep 19, 2021
Myanmar
May 14, 2023
Bata, Equatorial
Guinea
Mar 07, 2021
Goma, DR Congo
May 22, 2021
Disaster type
Earthquake
Armed conflict
Volcano eruption
Wildfire
Hurricane
Flood
Explosion


(a) Earthquake and volcano examples
RGB
SAR
GT
DamageFormer
MPF
Ours
Türkiye earthquake 00001065 512 × 512 px
noto earthquake 00000038 768 × 768 px
La Palma volcano 00000597 768 × 768 px
Background
Intact
Damaged
Destroyed


(b) Flood, conflict and wildfire examples
RGB
SAR
GT
DamageFormer
MPF
Ours
libya
flood
00000021
768 × 768
px
ukraine
conflict
00000354
512 × 512
px
hawaii
wildfire
00000003
512 × 512
px
Background
Intact
Damaged
Destroyed

Structure diagnostic panels

turkey-earthquake_00001065 | source crop [256, 0, 512, 512]

RGB



SAR



GT



Building B

Q: prior off

Q: prior on

b(Q)

H(Q)

Retrieval and correction diagnostics

Query: turkey-earthquake_00001065

RGB	SAR	GT	MPF	Cosine Reader	Supervised Reader	Rescued / harmed
						

Cosine-selected Train support



Support RGB



Support SAR



Train GT

ID:

Event:

Similarity:

Class / purity:

Supervised-selected Train support



Support RGB



Support SAR



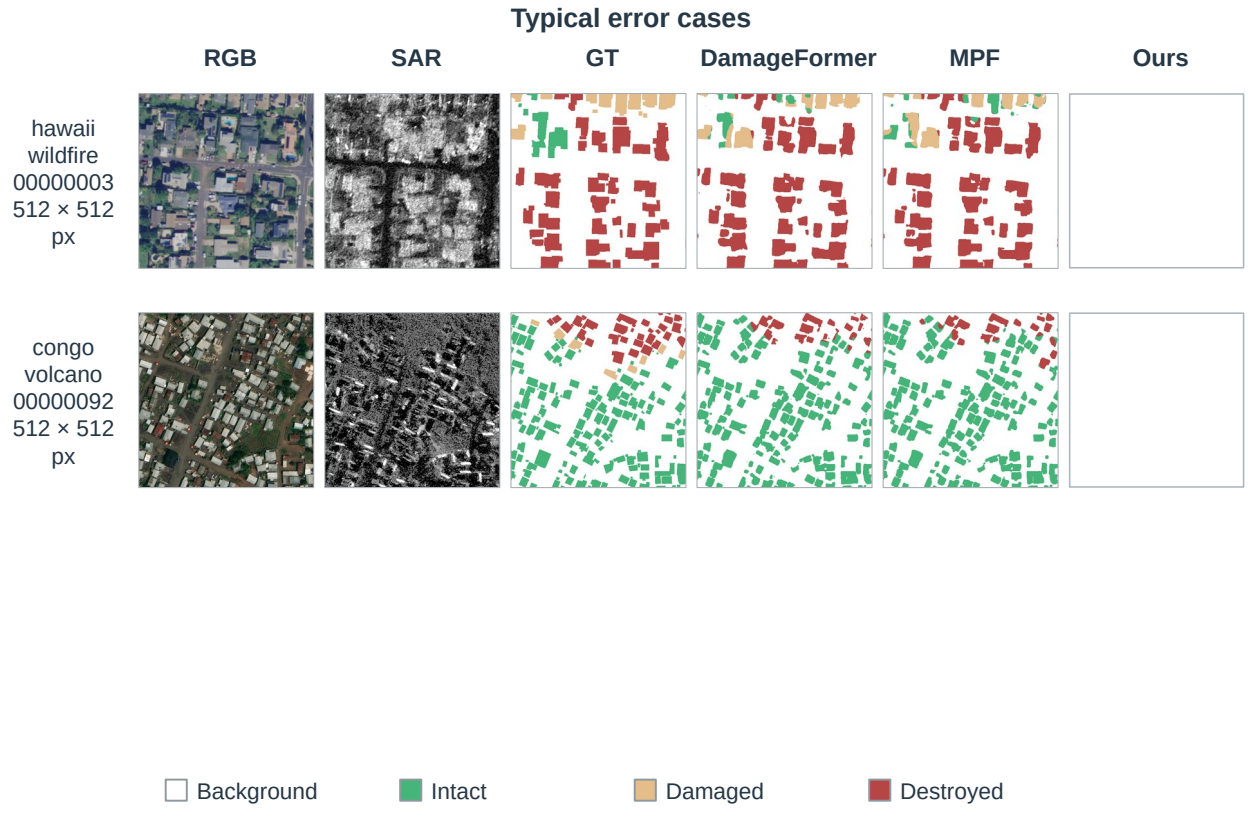
Train GT

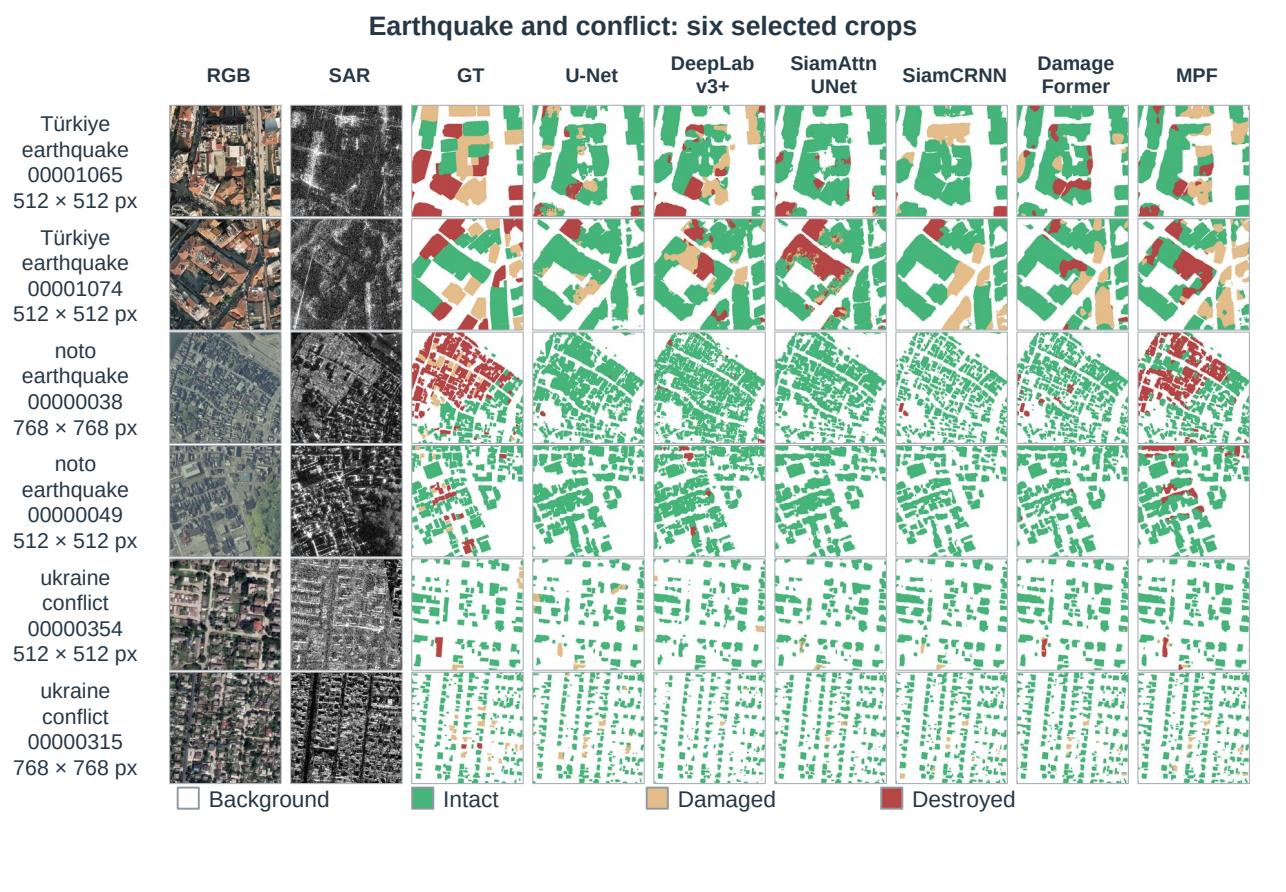
ID:

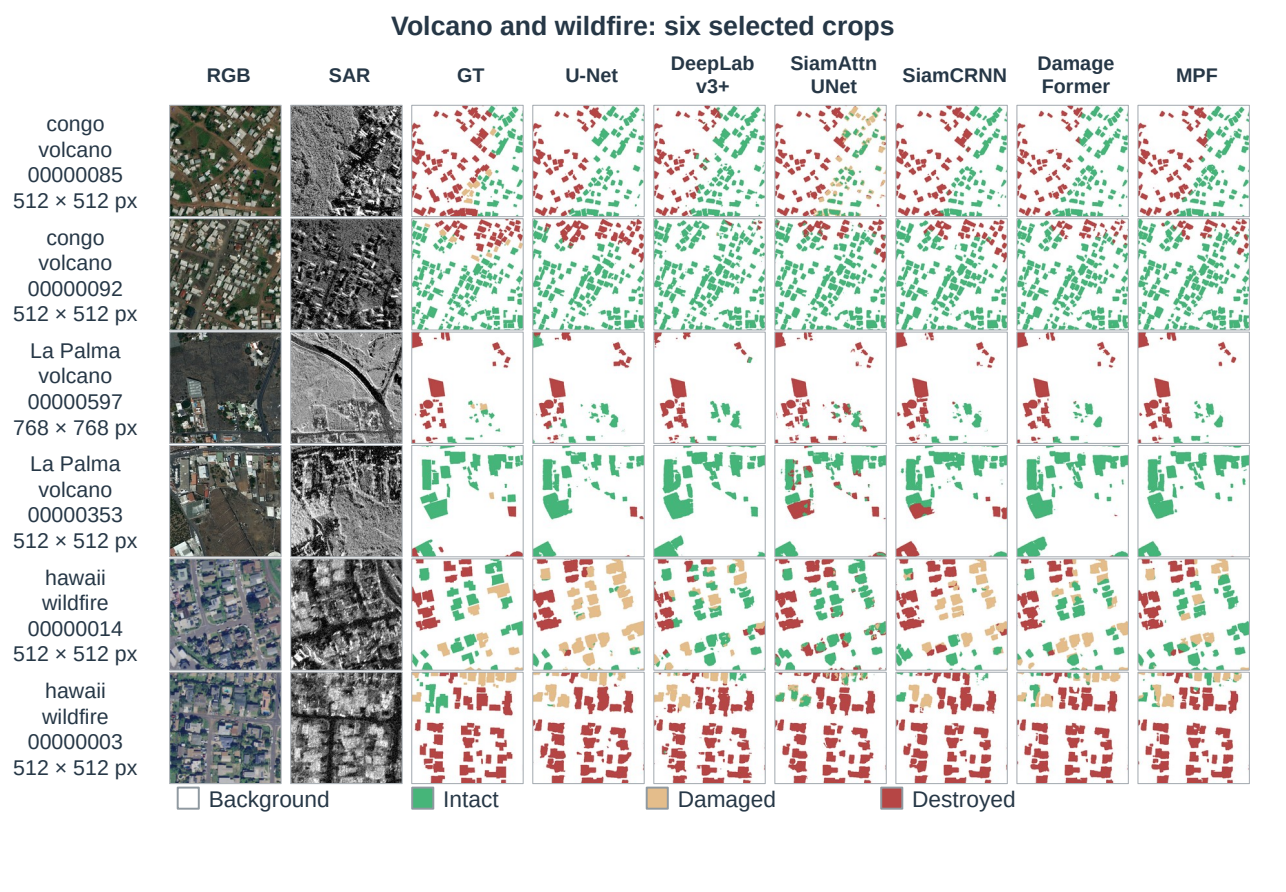
Event:

Similarity:

Class / purity:


Typical error cases
RGB
SAR
GT
DamageFormer
MPF
Ours
hawaii wildfire 00000003 512 × 512 px
congo volcano 00000092 512 × 512 px
Background
Intact
Damaged
Destroyed


Earthquake and conflict: six selected crops
RGB
SAR
GT
U-Net
DeepLab v3+
SiamAttn UNet
SiamCRNN
Damage Former
MPF
Türkiye earthquake 00001065 512 × 512 px
Türkiye earthquake 00001074 512 × 512 px
noto earthquake 00000038 768 × 768 px
noto earthquake 00000049 512 × 512 px
ukraine conflict 00000354 512 × 512 px
ukraine conflict 00000315 768 × 768 px
Background
Intact
Damaged
Destroyed


Volcano and wildfire: six selected crops
RGB
SAR
GT
U-Net
DeepLab v3+
SiamAttn UNet
SiamCRNN
Damage Former
MPF
congo volcano 00000085 512 × 512 px
congo volcano 00000092 512 × 512 px
La Palma volcano 00000597 768 × 768 px
La Palma volcano 00000353 512 × 512 px
hawaii wildfire 00000014 512 × 512 px
hawaii wildfire 00000003 512 × 512 px
Background
Intact
Damaged
Destroyed

Hurricane, flood and explosion: six selected crops

